

Instructions for Exam on course "Dynamical models for social behavior"

May 23, 2026

Exam Structure

The final exam consists of **two parts**:

1. Project Presentation

Each student is required to characterize one model from the list below through numerical simulations and analysis. The set of tasks provided is only a suggestion. Students are encouraged to explore the model in greater depth using additional sources.

Alternatively, with prior approval from the instructors, a student may choose and characterize another model of her/his interest.

You are expected to:

- Implement the model numerically;
- Run appropriate simulations and perform a characterization of the model;
- Prepare a short presentation (20 minutes) that includes:
 - Clear description of the model and its equations, and at least one example of a concrete application of the model;
 - Main parts of the implemented code;
 - Key plots, quantitative results and interpretations.

2. Theoretical Questions

There will be general questions on deterministic and stochastic processes referring to the topics covered during the lectures.

List of Exercises

1. **SIR Epidemiological Model** (Reference for the model: Chapter 10.2 of Murray, J.D. (2002). Mathematical Biology I: An Introduction (3rd ed.). Springer-Verlag, New York.)

The classic closed-population SIR model:

$$\begin{aligned}\frac{dS}{dt} &= -r SI, \\ \frac{dI}{dt} &= r SI - aI, \\ \frac{dR}{dt} &= aI,\end{aligned}$$

with S the susceptibles, who can catch the disease; I the infectives, who have the disease and can transmit it; and R the removed class, namely, those who have either had the disease, or are recovered, immune or isolated until recovered. $r > 0$ is the infection rate and $a > 0$ the removal rate of infectives. The total population is constant $N = S(t) + I(t) + R(t)$.

Possible Tasks: Integrate the equation of motions. Find some realistic numbers for values of r , a and consider different initial conditions for $S(0)$ and $I(0)$, with $R(0) = 0$. In the Murray's book you will find some possible example. Plot the time series of I , S , R , as well as I versus S . Discuss the difference between having an epidemic ($I(t) > I(0)$) or not. Discuss the time value where there is a peak in the infected.

2. **Random Walk on a Graph and PageRank algorithm** (Possible but not limiting References: Dobrow, Robert P. Introduction to stochastic processes with R. John Wiley & Sons, 2016)

Consider a graph (e.g. a small 6–8 nodes, either unweighted or weighted) with a meaningful real-world interpretation (e.g., citation network among research groups, flight routes between airports with passenger volumes as weights, or hyperlink graph of webpages).

Possible Tasks: Implement the evolution of the associated Markov chain via the transition matrix P . You can directly evolve the probability distribution to be in one node, as well as evolving a single agent that moves inside the graph. Compute the stationary probability. Expand on possible applications and additional implications, e.g. Google's Page Rank algorithm.

3. **Romeo and Juliet Model – Linear Dynamical System for Dyadic Romantic Interaction** (Reference: Maas, Han L.J. Complex Systems Research in Psychology, Chapter 4.3.2.1)

Consider the following linear system describing the evolution of feelings between Romeo ($R(t)$) and Juliet ($J(t)$):

$$\begin{aligned}\frac{dR}{dt} &= aR + bJ, \\ \frac{dJ}{dt} &= cR + dJ,\end{aligned}$$

where R and J represent the current level of love/feelings (positive = love, negative = hate), and the four parameters a, b, c, d define the “romantic style” of each person:

- $a > 0$: Romeo is encouraged by his own feelings;
- $a < 0$: Romeo retreats from his own feelings;
- $b > 0$: Romeo is encouraged by Juliet's feelings;
- $b < 0$: Romeo retreats from Juliet's feelings.

The same logic applies to Juliet with parameters c and d . Possible parameters sets are illustrated in the Maas' chapter.

Possible Tasks: Numerically integrate the system for the different parameters. Plot the time series $R(t)$ and $J(t)$. Draw the phase portrait in the (R, J) -plane, and analyse the long-term behaviour: convergence to a fixed point, damped oscillations, limit cycle, or divergence, depending on the signs of a, b, c, d . Perform a sensitivity analysis: vary one parameter at a time (e.g., change the strength of b or c) and discuss how the romantic style affects stability and final outcome.

4. **Van Geert Model – Coupled Development of Two Cognitive Abilities** (Reference: Maas, Han L.J. Complex Systems Research in Psychology, Chapter 4.3.3, based on van Geert (1991))

Consider the following coupled system describing the co-development of two interrelated cognitive abilities (e.g., vocabulary size X and syntactic complexity Y) in a child. Growth is limited by a shared capacity K , and each ability influences the growth rate of the other:

$$\begin{aligned}X_{t+1} &= (a + bY_t)X_t - \frac{aX_t^2}{K}, \\Y_{t+1} &= (c + dX_t)Y_t - \frac{cY_t^2}{K},\end{aligned}$$

Here $a > 0$ and $c > 0$ are the intrinsic growth rates, $K > 0$ is the carrying capacity, and b, d (typically negative) represent the interaction: one ability may support or suppress the growth of the other.

Realistic parameters calibrated to produce the two qualitatively different outcomes are shown in the Maas' chapter.

Possible Tasks: Implement the discrete iteration of the map. Plot the time series X_t and Y_t versus time for both parameter sets, as well as the phase portrait, by plotting directly X_t vs Y_t . Do it for different initial conditions. Analyze the long-term outcome: do the two abilities coexist and reach a joint equilibrium, or does one suppress the other? Identify the stable fixed points if possible. Perform a sensitivity analysis by varying the interaction parameters b and d (e.g., make them less negative or more negative) and the initial conditions. Discuss how the strength of the cross-influence affects whether the abilities develop together or one crowds out the other.

5. **Multiple Interacting Random Walkers with Hard-Core Exclusion** (Extension of the discrete 2D random walker – interacting particles on a lattice)

Consider a two-dimensional lattice with L^2 sites. Place N random walkers (particles) on distinct sites. The initial density is defined as $\rho = N/L^2$. At each discrete time step:

- Select one walker uniformly between the N ones at random.
- The selected walker attempts to jump in one of the four neighbour spaces, as done in class, with equal probability.
- The jump is allowed only if the target site is empty (hard-core exclusion); otherwise the walker stays in place.

Possible Tasks:

Implement the model and simulate the system for a sufficiently long time (e.g., 10^5 – 10^6 steps) using e.g., $L = 50$, with periodic boundary conditions. You will obtain for each particle i a trajectory $x_i(t)$, $y_i(t)$, with t the timestep of the simulation. For different particle densities $\rho = N/L^2$ (e.g., 0.1, 0.3, 0.5), compute the mean squared displacement (MSD) of the particles, defined as

$$\text{MSD}(t) = \langle (x_i(t) - x_i(0))^2 + (y_i(t) - y_i(0))^2 \rangle,$$

where the average is taken first over all N particles and then over independent realizations of the entire system. Plot $\text{MSD}(t)$ versus time for each density and compare the long-time diffusive behaviour with the case of a single non-interacting random walker. Generate snapshots of the

particle configurations at different times and discuss the possible formation of clusters and jamming effects at higher densities.

6. Discrete-Time Linear Birth-and-Death Chain – Population Dynamics and Extinction Risk (From Dobrow, Robert P. Introduction to Stochastic Processes With R, John Wiley & Sons 2016. Example 2.9 – Birth-and-death chain)

Consider a discrete-time birth-and-death Markov chain, with $\{N_t\}$ the population number, $t = 0, 1, 2, \dots$ and with the state space of population values $0, 1, 2, \dots$. The transition probabilities are given between a population value of i and one of j is given by

$$P_{i,j} = \begin{cases} q & \text{if } j = i - 1 \quad (\text{death}), \\ p & \text{if } j = i + 1 \quad (\text{birth}), \\ 1 - p - q & \text{if } j = i \quad (\text{stay}), \\ 0 & \text{otherwise,} \end{cases}$$

for all $i \geq 1$, with $0 \leq p, q$ and $p + q \leq 1$. In the case the population reaches $i = 0$, we have an absorbing state (extinction of the population), with $P_{0,0} = 1$ and $P_{0,j} = 0$ if $j \neq 0$.

Possible Tasks: Implement the Markov chain simulation, generating several independent trajectories. Start e.g. from $N_0 = 120$, and try different values of p and q ; simulate until absorption at 0 or up to a long horizon (e.g., $T = 500$ steps). Plot 10–15 representative trajectories of the population size N_t . Estimate numerically:

- the probability of eventual extinction (fraction of trajectories absorbed at 0);
- the mean time to extinction (only among trajectories that go extinct).

Compare the unbiased case $p = q$, with situations where p is either greater or lower than q . See possible differences changing the population size. Plot how extinction probability and mean extinction time change. Interpret the results.

7. Drift-Diffusion Model (DDM) – Evidence Accumulation for Speeded Decision Making (Possible References: Maas, Maas, Han L.J. Complex Systems Research in Psychology, Chapter 4.3.1; Physiological Review article for in deeper details)

The Drift-Diffusion Model is a classic stochastic accumulator model used to describe speeded two-alternative forced-choice tasks. The evidence $X(t)$ accumulates over time according to the stochastic differential equation

$$dX = I dt + \sigma dB_t,$$

where I is the drift rate (strength/quality of the stimulus evidence, $I > 0$ favours one alternative), $\sigma > 0$ is the noise intensity, and dB_t is the increment of a standard Brownian process. The process starts at $X(0) = 0$ (no initial bias). A decision is made as soon as the evidence reaches one of two absorbing boundaries: $+b$ (choose option 1) or $-b$ (choose option 2), with $b > 0$.

Possible parameters: drift rate $I = 0.1$ (strong evidence) or $I = 0.01$ (weak/difficult evidence); noise intensity $\sigma = 0.1$; boundary $b = 1$.

Possible Tasks: Implement the numerical simulation of many independent trials. Stop each trial when $X(t)$ reaches one of the two boundaries. Plot several representative trajectories of the evidence accumulation process (some that lead to correct and incorrect decisions). Compute and report:

- Accuracy (proportion of hits to the $+b$ boundary when $I > 0$);
- Mean response time (average time to absorption);
- Distribution of response times (histogram).

Analyze the speed–accuracy trade-off: repeat the simulations for different boundary separations b (e.g., 0.8, 1.0, 1.5) while keeping I and σ fixed. Plot mean response time versus accuracy. Perform an analysis by varying the drift rate I (different levels of stimulus difficulty) and discuss its effect on both speed and accuracy.

8. **Geometric Brownian Motion – Stochastic Model for Asset Prices and Multiplicative Growth** (Possible reference: Dobrow, Robert P. Introduction to Stochastic Processes With R, John Wiley & Sons 2016.)

The price S_t of an asset (stock, index, population size, etc.) follows the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dB_t,$$

where μ is the drift (expected growth rate), $\sigma > 0$ is the volatility, and dB_t is the variation of a Brownian process.

Possible parameters for a stock price (annual basis): - Initial price: $S_0 = 100$ - Drift: $\mu = 0.10$ (10% expected annual return) - Volatility: $\sigma = 0.20$ (20% annual volatility)

Given these parameters are on an annual basis, $T = 1$ corresponds to 1 year.

Possible Tasks: Implement the simulation and generate several independent paths. Plot representative price paths S_t (both on linear and logarithmic scale) and the average and variance as a function of time. The average should follow $E(S(t)) = S_0 e^{\mu t}$. At the final time T , plot the empirical distribution of S_T . It should be a log-normal distribution. Compute the probability over time to be at a value of S_t below the initial investment. Vary the drift μ and volatility σ and discuss how they affect the final distribution.

9. **Ornstein-Uhlenbeck Process – Mean-Reverting Model for Interest Rates and Asset Spreads** (Possible reference: Dobrow, Robert P. Introduction to Stochastic Processes With R, John Wiley & Sons 2016.)

In many financial applications, quantities such as short-term interest rates, credit spreads, or the difference (spread) between two cointegrated assets exhibit mean-reverting behavior. A classic model for such dynamics is the Ornstein-Uhlenbeck (OU) process (also known as the Vasicek model when applied to interest rates).

The process X_t (e.g., the interest rate r_t or the spread between two assets) satisfies the stochastic differential equation

$$dX_t = \kappa(\theta - X_t) dt + \sigma dB_t,$$

where $\kappa > 0$ is the speed of mean reversion, θ is the long-term mean, $\sigma > 0$ is the volatility, and dB_t is the increment of a standard Brownian motion.

Possible parameters (annual basis, for a short-term interest rate or a trading spread): Initial value $X_0 = 0.025$ (2.5%); Long-term mean $\theta = 0.04$ (4%); Mean reversion speed $\kappa = 0.4$; Volatility: $\sigma = 0.015$ (1.5% annual volatility).

Given these parameters are on an annual basis, $T = 1$ corresponds to 1 year.

Possible Tasks: Implement the simulation of the OU process and generate several independent paths. Plot representative paths of X_t (both on linear and, if appropriate, logarithmic scale). Plot the mean and variance as a function of time, along with theoretical mean $E[X_t] = X_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t})$. The sample mean should converge to the theoretical expression above. At a sufficiently large time T (e.g. $T = 5$ or $T = 10$), plot the empirical distribution of X_T and compare it to the theoretical stationary normal distribution, which should be a Gaussian distribution with mean θ and variance $\frac{\sigma^2}{2\kappa}$. Compute the probability over time that X_t exceeds a certain threshold (e.g. 6% for interest rates) or falls below the initial value. Vary the parameters κ , θ and σ and discuss how they affect e.g. the speed of mean reversion and the amplitude of fluctuations.